Supplementary Material for “Atomic and topological structure of domain wall networks in commensurate charge density waves of $2H$-$TaSe_2$”

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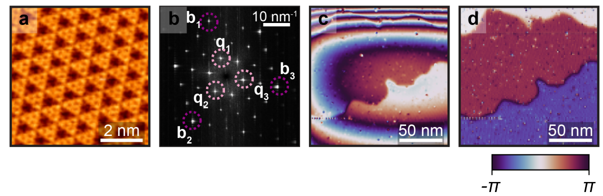

FIG. S1. Compensation of STM image distortion in spatial lock-in analysis. (a) Raw STM topography. (b) Fourier transform of the topography. (c) Raw phase map $\arg[Z_{\mathbf{q}_1}(\mathbf{r})]$ exhibiting spatial distortion. (d) Compensated phase map $\arg[Z_{\mathbf{q}_1}(\mathbf{r})] - \arg[Z_{3\mathbf{q}_1}(\mathbf{r})]/3$.

S1. VISUALIZATION OF DOMAIN WALL NETWORKS

S1.1. Extracting phase map of CDW from STM topography

S1.1.1. Spatial lock-in method

STM topography of $2H$-$TaSe_2$ with the 3×3 CDW can be decomposed into the Se atomic lattice term and the CDW term.

$$z^{\rm Raw}(\mathbf{r}) = z^{\rm Se}(\mathbf{r}) + z^{\rm CDW}(\mathbf{r}) \tag{S1}$$

Considering that the periodicities of the Se and CDW terms are $\mathbf{b}_i$ and $\mathbf{q}_i(= \mathbf{b}_i/3)$, respectively, we can define

$$z^{\rm Se}(\mathbf{r}) = \sum_{n=1}^{\infty}\sum_{i=1}^{3} z^{\rm Se}_{n,i} \cos[n\mathbf{b}_i \cdot \{\mathbf{r} - \mathbf{u}^{\rm Tip}(\mathbf{r})\}] \tag{S2}$$

$$z^{\rm CDW}(\mathbf{r}) = \sum_{n=1}^{\infty}\sum_{i=1}^{3} z^{\rm CDW}_{n,i} \cos[n\mathbf{q}_i \cdot \{\mathbf{r} - \mathbf{u}^{\rm Tip}(\mathbf{r})\} + \phi_i(\mathbf{r})] \tag{S3}$$

where i distinguishes three different FFT peaks $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ in Fig. S1 (b). $\mathbf{u}^{\rm Tip}(\mathbf{r})$ is an unwanted extrinsic distortion. $\phi_i(\mathbf{r})$ is the phase of CDW. Our primary goal is to find $\phi_i(\mathbf{r})$ from $z^{\rm Raw}(\mathbf{r})$.

Windowed Fourier transformation can be defined,

$$Z_{\mathbf{k}}(\mathbf{r}) = \int_{\mathbb{R}^2} d^2\mathbf{r}' z^{\rm Raw}(\mathbf{r}')\exp(-i\mathbf{k}\cdot\mathbf{r})w(\mathbf{r}-\mathbf{r}') \tag{S4}$$

where $w(\mathbf{r})$ is a window kernel centered at $\mathbf{r}$ that localizes the Fourier analysis to a neighborhood of that point. We used a normalized 2D Gaussian function.

$$w(\mathbf{r}) = \frac{1}{2\pi\sigma^2} \exp\left(\frac{-|\mathbf{r}|^2}{2\sigma^2}\right) \quad \text{(S5)}$$

σ should be finely tuned considering the trade-off between the real-space resolution required to resolve highly localized structures, such as domain walls, and the momentum-space resolution needed to isolate specific wave vectors. If we insert $\mathbf{q}_1$ into $Z_{\mathbf{k}}(\mathbf{r})$ then the result is

$$Z_{\mathbf{q}_i}(\mathbf{r}) \simeq (z_{1,i}^{\mathbf{CDW}}/2)\exp[i\{\phi_i(\mathbf{r}) - \mathbf{q}_i \cdot \mathbf{u}^{\mathrm{Tip}}(\mathbf{r})\}] \quad \text{(S6)}$$

However, due to the distortion $\mathbf{u}^{\mathrm{Tip}}(\mathbf{r})$, $\arg[Z_{\mathbf{q}_i}(\mathbf{r})]$ does not yield $\phi_i(\mathbf{r})$ directly; instead, it provides a distorted phase, $\phi_i(\mathbf{r}) - \mathbf{q}_i \cdot \mathbf{u}^{\mathrm{Tip}}(\mathbf{r})$.

S1.1.2. Artifact compensation

Figure S1(c) shows the distorted phase map of $\phi_1(\mathbf{r})$. While the distortion originating from the tip drift is negligible within a small scan range, it becomes significant in our large-scale measurements. Fortunately, we can compensate for it using the fact that the corrugation of Se atoms has the same distortion.

$$\begin{aligned} Z_{3\mathbf{q}_i}(\mathbf{r}) \simeq & \frac{z_{1,i}^{\mathrm{Se}}}{2} \exp[i\{-3\mathbf{q}_i \cdot \mathbf{u}^{\mathrm{Tip}}(\mathbf{r})\}] \\ & + \frac{z_{3,i}^{\mathrm{CDW}}}{2} \exp[i\{\phi_i(\mathbf{r}) - 3\mathbf{q}_i \cdot \mathbf{u}^{\mathrm{Tip}}(\mathbf{r})\}] \\ \simeq & \frac{z_{1,i}^{\mathrm{Se}}}{2} \exp[i\{-3\mathbf{q}_i \cdot \mathbf{u}^{\mathrm{Tip}}(\mathbf{r})\}] \end{aligned} \quad \text{(S7)}$$

the second $\simeq$ is justified by $z_{1,i}^{\mathrm{Se}} \gg z_{3,i}^{\mathrm{CDW}}$. We obtained artifact-considered phase map

$$\phi_i(\mathbf{r}) \simeq \arg[Z_{\mathbf{q}_i}(\mathbf{r})] - \arg[Z_{3\mathbf{q}_i}(\mathbf{r})]/3 \quad \text{(S8)}$$

Figures S1(c) and S1(d) show the efficacy of this compensation.

S1.2. Order parameter space

S1.2.1. Phase map (ϕ_1, ϕ_2, ϕ_3)

The obtained phase map $\phi_i(\mathbf{r})$ clearly shows the phase lock-in in Fig. S2. Each $\phi_i(\mathbf{r})$ can have three cyclic phases $\{-2\pi/3, 0, 2\pi/3\}$. These phases ϕ_i are not independent but satisfy the relation

$$\phi_1(\mathbf{r}) + \phi_2(\mathbf{r}) + \phi_3(\mathbf{r}) = 2\pi/3 \quad \text{(S9)}$$

This value determines the CDW structure: hollow-centered ($= 2\pi/3$), cation-centered ($= -2\pi/3$), or anion-centered ($= 0$).

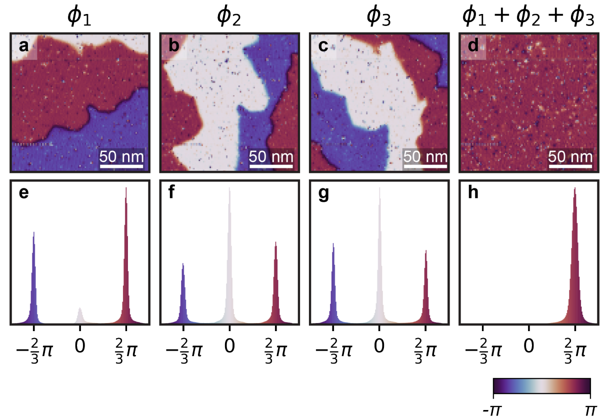

FIG. S2. Phase maps and corresponding phase histograms. (a–d) Phase maps of $\phi_1(\mathbf{r})$, $\phi_2(\mathbf{r})$, $\phi_3(\mathbf{r})$, and the sum $\phi_1(\mathbf{r}) + \phi_2(\mathbf{r}) + \phi_3(\mathbf{r})$, respectively. (e–h) Phase histograms corresponding to the phase maps in (a–d).

S1.2.2. $(\tilde{n}_1, \tilde{n}_2)$ and t representation

All possible combinations of (ϕ_1, ϕ_2, ϕ_3) are $3 \times 3 \times 3 = 27$, but the hollow-centered CDW constraints reduce this number to nine [Table S1]. In our main text, to emphasize the translational symmetry breaking, nine ground states are denoted in the (n_1, n_2) notation and the exact mapping of (ϕ_1, ϕ_2, ϕ_3) is shown in Table S1, with their relationship visualized in Fig. S3. A continuous mapping between $\phi_1, \phi_2, \phi_3 \in [-\pi, \pi)$ and $\tilde{n}_1, \tilde{n}_2 \in [-1.5, 1.5)$, $t \in [-0.5, 0.5)$ can be defined by

$$\begin{bmatrix}\phi_1\\ \phi_2\\ \phi_3\end{bmatrix} = \frac{2}{3}\pi \left\{ \begin{bmatrix} 0 & -1 & 1\\ 1 & 0 & 1\\ -1 & 1 & 1\end{bmatrix} \begin{bmatrix}\tilde{n}_1\\ \tilde{n}_2\\ t\end{bmatrix} + \begin{bmatrix}1\\0\\0\end{bmatrix} \right\} \quad \text{(S10)}$$

$\tilde{n}_1$ and $\tilde{n}_2$ are the continuous analogs of n_1 and n_2, respectively, and the extra coefficient t was introduced for the mapping to ensure a bijective mapping.

In our case, t is almost 0 because of the constraint $\phi_1(\mathbf{r}) + \phi_2(\mathbf{r}) + \phi_3(\mathbf{r}) = 2\pi/3$.

(n_1, n_2)	(ϕ_1, ϕ_2, ϕ_3)
$(0, 0)$	$(2\pi/3, 0, 0)$
$(+, 0)$	$(2\pi/3, 2\pi/3, -2\pi/3)$
$(0, +)$	$(0, 0, 2\pi/3)$
$(-, -)$	$(-2\pi/3, -2\pi/3, 0)$
$(+, -)$	$(-2\pi/3, 2\pi/3, 2\pi/3)$
$(-, 0)$	$(2\pi/3, -2\pi/3, 2\pi/3)$
$(0, -)$	$(-2\pi/3, 0, -2\pi/3)$
$(+, +)$	$(0, 2\pi/3, 0)$
$(-, +)$	$(0, -2\pi/3, -2\pi/3)$

TABLE S1. Correspondence between (n_1, n_2) and (ϕ_1, ϕ_2, ϕ_3) notations.

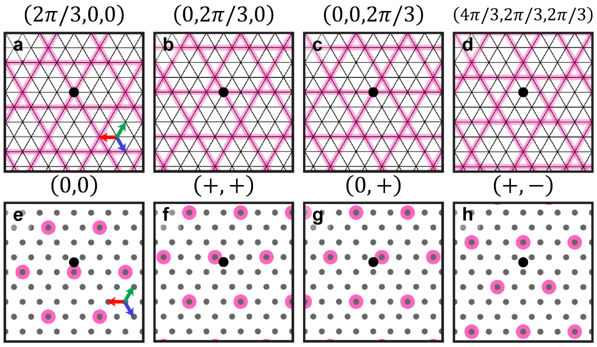

FIG. S3. Visualized correspondence between the (n_1, n_2) and (ϕ_1, ϕ_2, ϕ_3) configurations. (a–d) Lattice representation where the black filled circle represents the origin point, and the red, green, and blue arrows correspond to the three basis vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$. Black and pink lines parallel to $\mathbf{a}_i$ indicate the maxima of the $\mathbf{b}_i$ and $\mathbf{q}_i$ components, respectively, and pink lines reflect the relative phase (ϕ_1, ϕ_2, ϕ_3) denoted above each panel. (e–f) Superposition model where gray dots represent the lattice sites and the pink circle denotes the maximum of the CDW cluster from the superposition of $\mathbf{q}_1$, $\mathbf{q}_2$, and $\mathbf{q}_3$. The indices (n_1, n_2) indicate the position of the center of the CDW cluster.

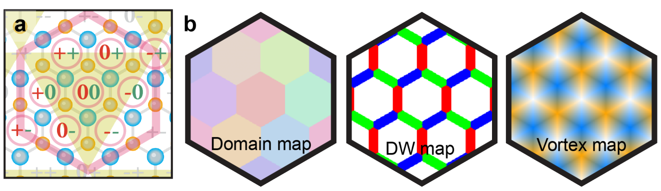

FIG. S4. Color maps for domains, domain walls (DWs), and vortices. (a) Schematic of the order parameter space. (b) Color maps used to extract topological features. The domain map distinguishes the nine ground states, the DW map distinguishes r, g, and b DWs using red, green, and blue, respectively, and the vortex map distinguishes rgb (sky-blue) and $r^{-1}g^{-1}b^{-1}$ (orange) vortex core states.

S1.3. Capturing topological defects

While $\phi_1(\mathbf{r})$, $\phi_2(\mathbf{r})$, and $\phi_3(\mathbf{r})$ (or $\tilde{n}_1$, $\tilde{n}_2$, t) contain comprehensive information regarding the domain wall network (DWN) in the system, they are individually unsuitable for rendering the global topology. To address this, we integrated these phase fields to directly visualize the spatial distribution of domains, DWs, and vortices by introducing a colormap function $\mathcal{C}(\phi_1, \phi_2, \phi_3)$ that yields a specific color vector (R, G, B) for each spatial position. To systematically visualize the spatial distribution of these topological features, we introduce three complementary colormap functions $\mathcal{C}_d$, $\mathcal{C}_{dw}$, and $\mathcal{C}_{vor}$. The colormap functions are visualized in Fig. S4, and their rigorous formalizations are described below.

S1.3.1. Domain map $\mathcal{C}_d$

The colormap function $\mathcal{C}_d(\tilde{n}_1, \tilde{n}_2)$ yields one of the nine color vectors $c_{(n_1,n_2)}$ that minimizes $|(\tilde{n}_1 - n_1)\mathbf{a}_1 + (\tilde{n}_2 - n_2)\mathbf{a}_2|$.

S1.3.2. DW map $\mathcal{C}_{dw}$

The function $\mathcal{C}_{dw}(\tilde{n}_1, \tilde{n}_2)$ is introduced to distinguish between r, g, and b DWs. This is because an r DW connects two adjacent domains, (n_1, n_2) and (n_1+1, n_2), its phase center is naturally located at $(n_1 + 1/2, n_2)$. Similarly, the phase centers for g and b DWs are assigned to $(n_1, n_2 + 1/2)$ and $(n_1 + 1/2, n_2 + 1/2)$, respectively. To map these features from continuous phase coordinates $(\tilde{n}_1, \tilde{n}_2)$, we define an appropriate neighborhood around each DW center. Let c_1, c_2, and c_3 denote the colors assigned to r, g, and b DWs, respectively, and c_0 denote the color of the domains. The colormap function $\mathcal{C}_{dw}$ is then defined as:

$$\mathcal{C}_{dw}(\tilde{n}_1, \tilde{n}_2) = \begin{cases} c_i & \text{if } (\tilde{n}_1, \tilde{n}_2) \in \mathcal{R}_i \quad (i = 1, 2, 3), \\ c_0 & \text{otherwise}, \end{cases} \tag{S11}$$

where the DW region $\mathcal{R}_i$ is defined as:

$$\mathcal{R}_i = \Bigg\{ (\tilde{n}_1, \tilde{n}_2) \;\Bigg|\; \tilde{n}_1\mathbf{a}_1 + \tilde{n}_2\mathbf{a}_2 = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + \frac{1}{2}\mathbf{a}_i + s_i\left(\frac{1}{6} - \delta\right)\mathbf{a}_i + s_{i+1}\delta\mathbf{a}_{i+1} + s_{i-1}\delta\mathbf{a}_{i-1}, \text{ for } n_1, n_2 \in \{0, 1, 2\} \text{ and } s_1, s_2, s_3 \in [-1, 1) \Bigg\} \tag{S12}$$

Here, δ is a small number used to control the thickness of the DW in the phase map.

S1.3.3. Vortex map $\mathcal{C}_{vor}$

$\mathcal{C}_{vor}(\tilde{n}_1, \tilde{n}_2)$ is a colormap designed to distinguish the two types of vortices (rgb and $r^{-1}g^{-1}b^{-1}$) based on the center position of the CDW. The emergent state near the vortex core is cation-centered for rgb and anion-centered for $r^{-1}g^{-1}b^{-1}$. Using $(\tilde{n}_1, \tilde{n}_2)$, vortex core states can be represented as $n_1\mathbf{a}_1 + n_2\mathbf{a}_2 \pm \frac{1}{3}(\mathbf{a}_i - \mathbf{a}_{i+1})$, where $+$ represents the anion-centered state and $-$ represents the cation-centered state. For the colors of the hollow-, anion-, and cation-centered structures c_H, c_A, and c_C, we can define $\mathbf{c}_{\text{vor}}(s_A, s_C) = s_A\mathbf{c}_A + s_C\mathbf{c}_C + (1 - s_A - s_C)\mathbf{c}_H$. We can define s_A and s_C for $(\tilde{n}_1, \tilde{n}_2)$ by

$$\tilde{n}_1\mathbf{a}_1 + \tilde{n}_2\mathbf{a}_2 = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + \frac{s_A}{3}(\mathbf{a}_i - \mathbf{a}_{i+1}) + \frac{s_C}{3}(\mathbf{a}_{j+1} - \mathbf{a}_j) \tag{S13}$$

where $i \neq j$.

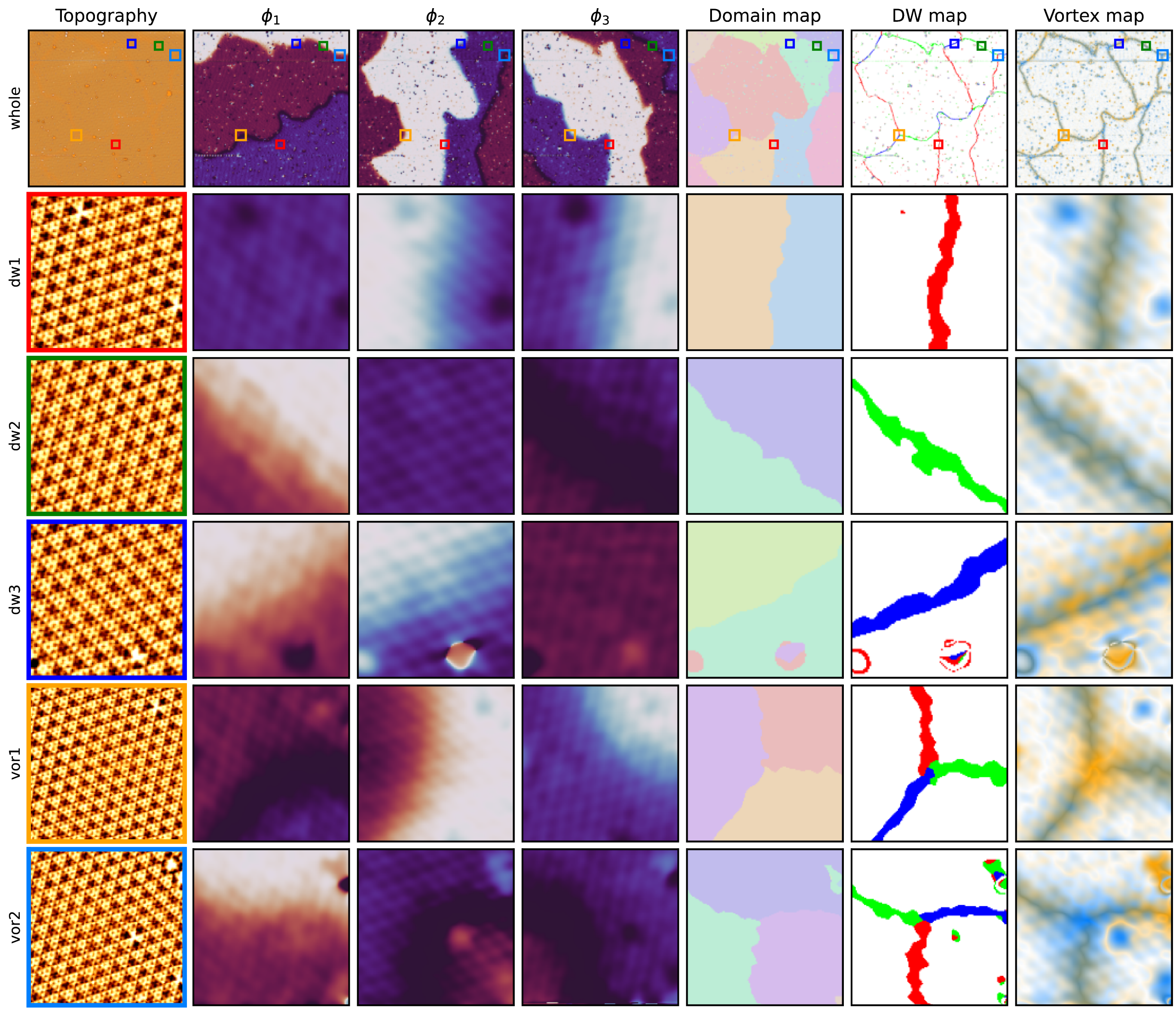

FIG. S5. Spatial mapping and extraction of topological features from raw STM data. The first column displays the reference STM topography, where the top panel contains five sub-regions outlined by colored boxes, as detailed in the subsequent rows: red box for "dw1" (r DW), green box for "dw2" (g DW), purple box for "dw3" (b DW), orange box for "vor1" (rgb vortex), and sky-blue box for "vor2" ($r^{-1}g^{-1}b^{-1}$ vortex). The second, third, and fourth columns present the extracted phase components ϕ_1, ϕ_2, and ϕ_3, respectively. The fifth, sixth, and seventh columns present the domain map, DW map, and vortex map, which are constructed using the colormap functions $\mathcal{C}_d$, $\mathcal{C}_{dw}$, and $\mathcal{C}_{vor}$ defined in Section S1.3, respectively. The red and blue dashed circles in the "dw3" row indicate local point defects or pinning centers along the domain wall that host localized phase perturbations.

S2. TOPOLOGICAL NATURE OF DOMAIN WALL NETWORKS

S2.1. Proper boundary problem

We first define the boundary charge $\Phi(L)$ for a given DWN. The boundary charge $\Phi(L)$ is represented as a sequence constructed from the phase shifts $\{r, g, b\}$ and their inverses $\{r^{-1}, g^{-1}, b^{-1}\}$, corresponding to the phase shifts encountered upon traversing a clockwise loop L in real space. A fundamental question arises: can an arbitrary sequence of phase shifts serve as a valid boundary of a physical DWN?

S2.1.1. Phase accumulation condition

To represent a physical boundary, the sequence must satisfy certain constraints. The first condition is the

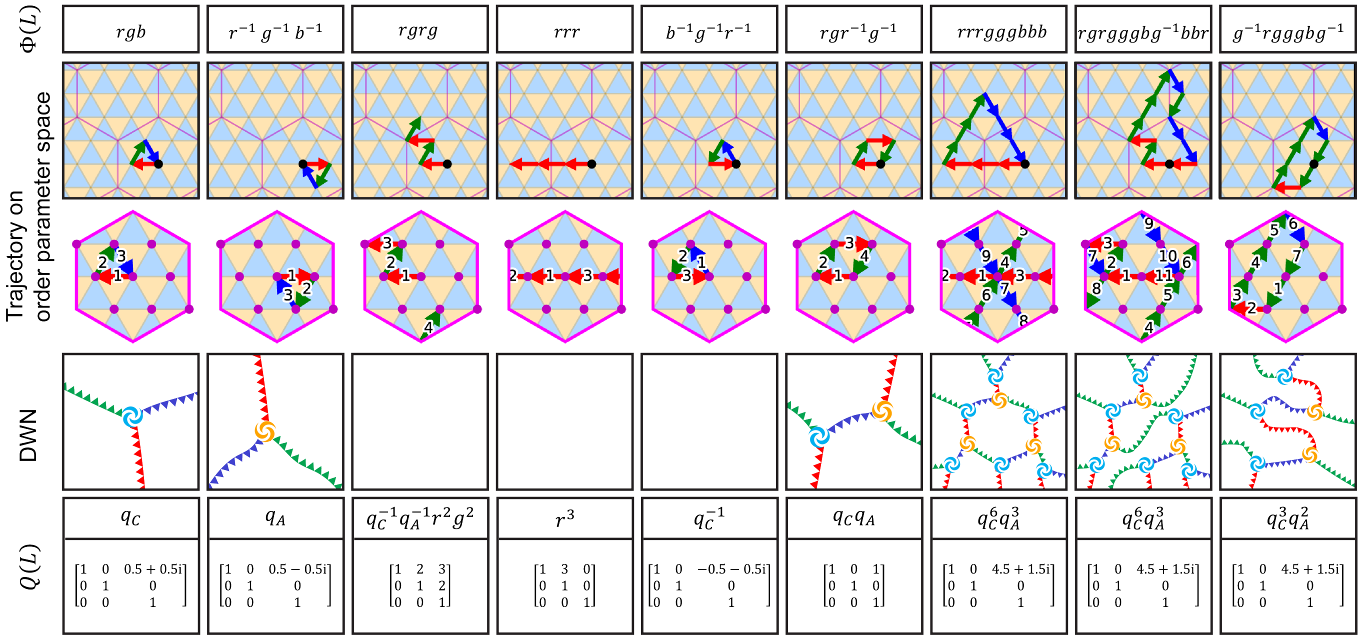

FIG. S6. Schematic illustration of the proper boundary problem in the domain wall network, showcasing paths that enclose different configurations of topological defects. The figure is structured into five rows: The first row displays the boundary charge $\Phi(L)$ at the top of each column. The second and third rows show the trajectories of the boundary charges in the order parameter space under the extended and folded schemes, respectively. The fourth row shows the corresponding real-space DWN configurations, where columns 3–5 represent examples of paths that do not form physically allowed DWN structures (leaving their panels blank). The fifth row displays the net bulk topological charge $Q(L)$ along with its corresponding 3×3 matrix representation in the Heisenberg group.

phase accumulation condition, which dictates that the net accumulated phase shift along the loop must be zero; geometrically, this means that the trajectory of the CDW phase in the order parameter space must form a closed loop. For example, a boundary represented by the sequence $rgrg$ is physically forbidden because its accumulated phase shift is $(+,0)+(0,+)+(+,0)+(0,+)=(-,-)\neq(0,0)$, which corresponds to an open trajectory in the order parameter space [Fig. S6, third column]. The phase accumulation condition can be expressed as

$$n_r \equiv n_g \equiv n_b \pmod 3 \tag{S14}$$

where n_i represents the net count of phase shifts of type i (with inverse elements counted negatively). The two vortices, rgb and $r^{-1}g^{-1}b^{-1}$, naturally satisfy this condition. However, this phase accumulation condition is necessary but not sufficient to ensure a valid boundary. For instance, the sequence rrr [Fig. S6, fourth column] yields an accumulated phase shift of $(3,0)\equiv(0,0) \pmod 3$ and forms a closed trajectory in the order parameter space, yet it does not represent a physically allowed boundary.

S2.1.2. Vortex decomposability

Physically, any closed boundary loop in the DWN must be decomposable into a combination of physically allowed vortices. In the commensurate CDW phase of $2H$-$TaSe_2$, the only physically allowed vortices are the rgb and $r^{-1}g^{-1}b^{-1}$ vortices. For example, the boundary sequence $rgr^{-1}g^{-1}$ is vortex-decomposable because it can be algebraically decomposed as

$$rgr^{-1}g^{-1} = (rgb)\cdot(b^{-1}r)(r^{-1}g^{-1}b^{-1})(b^{-1}r)^{-1} \tag{S15}$$

representing a combination of one rgb vortex and one $r^{-1}g^{-1}b^{-1}$ vortex. Here, the conjugation terms $(b^{-1}r)...(b^{-1}r)^{-1}$ physically describe the configuration of the DWs connecting the vortices. More complex boundaries, such as $g^{-1}rgggbg^{-1}$, can be similarly decomposed.

$$\begin{aligned} g^{-1}rgggbg^{-1} = {} & g^{-1}(rgb)g\cdot r(r^{-1}g^{-1}b^{-1})r^{-1} \\ & \cdot(rgb)\cdot b^{-1}(r^{-1}g^{-1}b^{-1})b \\ & \cdot g(rgb)g^{-1}. \end{aligned} \tag{S16}$$

We can generalize the decomposition form

$$\prod_{i=1}^{n} l_i \tilde{q}_i l_i^{-1} \tag{S17}$$

where $\tilde{q}_i$ is an rgb or $r^{-1}g^{-1}b^{-1}$ vortex and l_i is a sequence of phase shifts that connects the vortices.

This algebraic representation of the DWN topology is equivalent to a van Kampen lemma for the word problem in geometric group theory [1]. From (S17), if $\Phi(L)$ is vortex-decomposable, it must satisfy the condition

$$n_r = n_g = n_b \tag{S18}$$

The conjugation parts, l_i and l_i^{-1}, cancel each other out, leaving only the $\tilde{q}_i$ terms, satisfying (S18). Generally, vortex decomposability includes the phase accumulation condition since the individual vortices naturally satisfy the phase accumulation rule.

However, the condition $n_r = n_g = n_b$ is still not sufficient. For example, the sequence $b^{-1}g^{-1}r^{-1}$ (anti-vortex of the rgb vortex) satisfies $n_r = n_g = n_b = -1$, but the anti-vortex is not physically allowed. Geometrically, vortex decomposability can be summarized as follows: the trajectory in the extended-scheme order parameter space must be closed (S18) and clockwise (excluding anti-vortex cases).

S2.2. Boundary-charge correspondence

S2.2.1. Bulk topological charge

While the exact decomposition form (S17) contains the complete topological information of the DWN, it is often too complex for practical calculations. To simplify this, we can substitute $\tilde{q}_i$ with the abelian element q_i^{Ab} and define the bulk charge

$$\begin{aligned} Q(L) &= \prod_{i=1}^{n} l_i q_i^{\mathrm{Ab}} l_i^{-1} \\ &= \prod_{i=1}^{n} q_i^{\mathrm{Ab}} l_i l_i^{-1} = \prod_{i=1}^{n} q_i^{\mathrm{Ab}} \\ &= q_C^{n_C} q_A^{n_A} \end{aligned} \tag{S19}$$

Although this simplification loses some detailed information regarding the specific attachment paths of the DWs, it preserves essential topological information such as the vortex decomposability and the number of internal vortices. For example, although the seventh and eighth columns of Fig. S6 represent highly distinct real-space configurations with boundary charges $rrrgggbbb$ and $rgrgggbg^{-1}bbr$, they yield the same bulk charge $Q(L) = q_C^6 q_A^3$, enclosing the identical number of vortices (six rgb and three $r^{-1}g^{-1}b^{-1}$ vortices).

S2.2.2. Charge reduction process

Although we can determine the bulk charge by finding the decomposition form (S17) and abelianizing it, this process is too complicated. The reduction rules are as follows:

$$\begin{aligned} &\text{Rule 1.} \quad q_C = rgb \\ &\text{Rule 2.} \quad q_A = r^{-1}g^{-1}b^{-1} \\ &\text{Rule 3.} \quad q_C \text{ and } q_A \text{ are abelian} \end{aligned} \tag{S20}$$

These can be rewritten as:

$$\begin{aligned} &\text{Rule 1.} \quad b = q_C r^{-1} g^{-1} \\ &\text{Rule 2.} \quad q_C q_A = [r,g] \\ &\text{Rule 3.} \quad q_C \text{ and } q_A \text{ are abelian} \end{aligned} \tag{S21}$$

where $[r,g] = r^{-1}g^{-1}rg$. (S20) and (S21) are equivalent, but (S20) more clearly reflects our original physical idea, whereas (S21) is more practical for algebraic calculations. Now, we show an example, $g^{-1}rgggbg^{-1}$ (the last column of Fig. S6), using (S21):

$$\begin{aligned} g^{-1}rgggbg^{-1} &= g^{-1}rggg(q_C g^{-1}r^{-1})g^{-1} \quad \text{(Rule 1.)} \\ &= q_C g^{-1}rgggg^{-1}r^{-1}g^{-1} \\ &= q_C g^{-1}rggr^{-1}g^{-1} \\ &= q_C g^{-1}(rgg)r^{-1}g^{-1} \\ &= q_C g^{-1}(ggr[r,g]^2)r^{-1}g^{-1} \\ &= q_C g^{-1}(ggrq_C^2 q_A^2)r^{-1}g^{-1} \quad \text{(Rule 2.)} \\ &= q_C^3 q_A^2 g^{-1}ggrr^{-1}g^{-1} \\ &= q_C^3 q_A^2 \end{aligned} \tag{S22}$$

The entire process can be summarized in two steps: First, eliminate b using Rule 1, and then reorder r and g using Rule 2.

S2.2.3. Heisenberg group representation

To provide a more straightforward and concrete framework for the charge reduction process, we can map the phase shifts and the topological charges onto the Heisenberg group $\mathcal{H}$. For the phase shifts:

$$r,g,b \mapsto \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 & 0.5+0.5i \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \tag{S23}$$

And for the topological charges:

$$q_C, q_A \mapsto \begin{bmatrix} 1 & 0 & 0.5+0.5i \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0.5-0.5i \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{S24}$$

These representations satisfy (S21). If $\Phi(L)$ is a closed path, it reduces to a matrix of the form:

$$\Phi(L) \mapsto \begin{bmatrix} 1 & 0 & z_q \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{S25}$$

As shown above, if the resulting matrix has zero elements at $(1,2)$ and $(2,1)$, the result can be represented as $q_C^{n_C} q_A^{n_A}$, where the exponents of the charge are determined by:

$$\begin{aligned} n_C &= \mathrm{Re}(z_q) + \mathrm{Im}(z_q) \\ n_A &= \mathrm{Re}(z_q) - \mathrm{Im}(z_q) \end{aligned} \tag{S26}$$

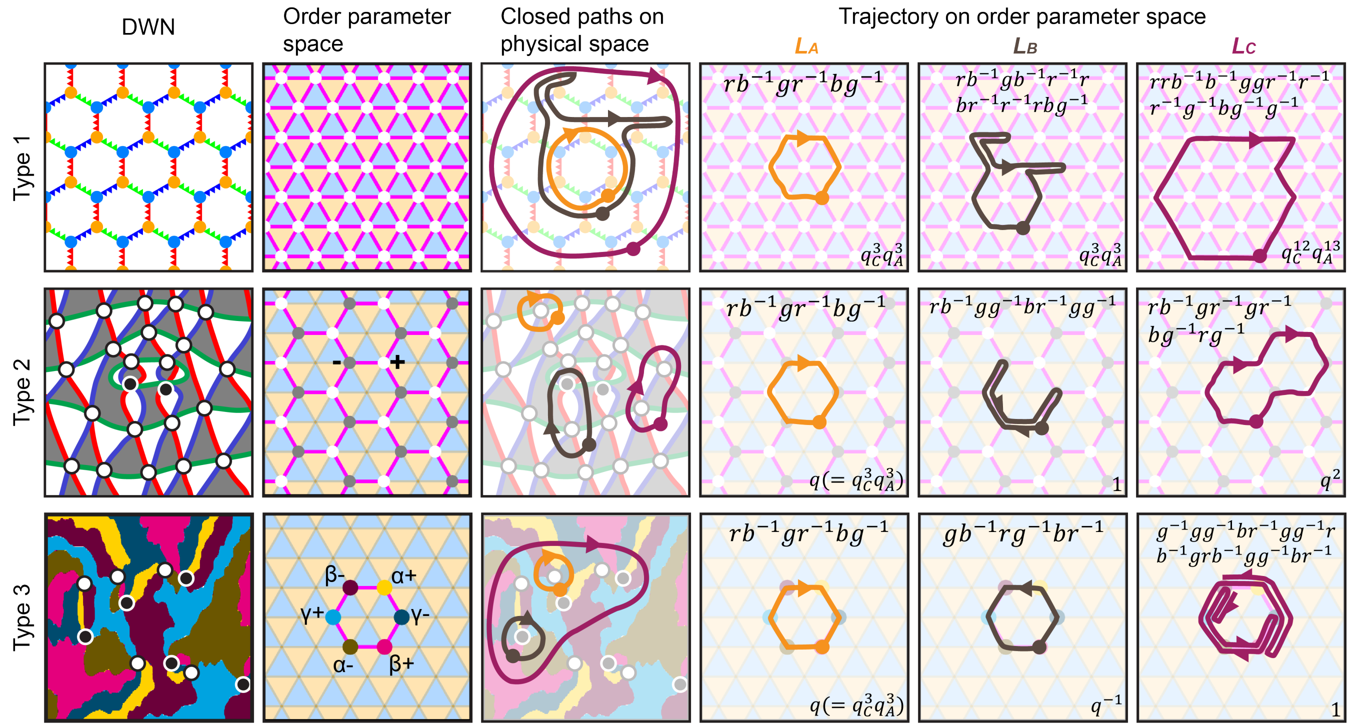

FIG. S7. Application of the topological charge framework to various two-dimensional domain wall networks (DWNs) with different geometric and topological symmetries, categorized into three types. Each row represents a specific DWN configuration: Type 1 (honeycomb DWN with Z_3 vortices), Type 2 (alternating triangular DWN with Z_6 vortices), and Type 3 (DWN with sixfold domain states and Z_6 vortices). The first column displays the real-space schematics of the DWNs, and the second column shows their corresponding order parameter spaces. The third column presents arbitrary closed paths on the physical space. The fourth to sixth columns show the corresponding trajectories in the order parameter space for the loops L_A (orange), L_B (brown), and L_C (magenta), respectively. For each trajectory panel, the algebraic expression at the top represents the boundary charge $\Phi(L)$, while the bulk charge $Q(L)$ is shown at the bottom, demonstrating the general applicability of the boundary-charge correspondence.

S2.3. Application to other DWNs

The applicability of this topological framework is not limited to $2H$-TaSe$_2$. We categorize several DWNs into three types based on their geometric structures and vortex types.

S2.3.1. *Type 1: Z_3 vortices in a honeycomb DWN*

Type 1 DWN is characterized by the honeycomb network. All domains are surrounded by six DWs. Three types of DWs (r, g, b) and two types of vortices (rgb and $r^{-1}g^{-1}b^{-1}$) exist in the system. The CCDW in $2H$-TaSe$_2$ (this study), the nearly commensurate CDW in $1T$-TaS$_2$ [2], and marginally twisted TMDs in anti-parallel (AP) stacking [3–5] are examples. In the order parameter space, each phase has six neighboring phases directly connected by the phase shifts $r, g, b, r^{-1}, g^{-1}, b^{-1}$. Two alternating triangular loops in the order parameter space represent rgb (q_C) and $r^{-1}g^{-1}b^{-1}$ (q_A) vortices. The proper boundary condition and boundary-charge correspondence are the same as those established above.

S2.3.2. *Type 2: Z_6 vortices in alternating triangular DWN*

Type 2 is characterized by the alternating triangular DWN with Z_6 vortices. In contrast to Type 1, each domain has only three adjacent domains and forms a $rb^{-1}gr^{-1}bg^{-1}$ vortex structure. Two types of domains, + and −, exist, represented by white and gray regions in Fig. S7. In the order parameter space, a + (−) domain neighbors only three − (+) domains, connected by r, g, b (r^{-1}, g^{-1}, b^{-1}) phase shifts. In Type 2, the sequence of exponents in $\Phi(L)$ must alternate between + and −, i.e., +, − alternatingly. For example, the exponent of $\Phi(L_{2A}) = rb^{-1}gr^{-1}bg^{-1}$ is $+-+-+-$, which satisfies the condition. In contrast, the exponent of $\Phi(L_{1B}) = rb^{-1}gb^{-1}r^{-1}rbr^{-1}r^{-1}rbg^{-1}$ is $+-+--++--++-$, which violates the condition.

Although the condition for a proper boundary is different, we do not require a new charge reduction rule. For $\Phi(L_{1A}) = rb^{-1}gr^{-1}bg^{-1}$, the resulting bulk charge is $q_C^3 q_A^3$. We introduce a new charge symbol, q =

$q_C^3 q_A^3$, to represent the Z_6 vortex. We can check that the boundary-charge correspondence works well for all boundary examples in Fig. S7. L_{2B} contains one vortex and one anti-vortex. Their trajectory is trivial, and the resulting bulk charge is 1. This implies that a vortex and an anti-vortex can annihilate each other in this system. L_{2C} contains two vortices, and their trajectory encloses these two vortices, yielding a bulk charge of q^2. The examples of Type 2 are $2H$-$NbSe_2$ [6], marginally twisted bilayer graphene [7, 8], and marginally twisted TMDs in parallel stacking [3, 4].

S2.3.3. Type 3: Z_6 vortices in a sixfold DWN

Type 3 is characterized by the DWN with Z_6 vortices and six domain states. The vortex structure is the same as in Type 2, but DWN forms an irregular pattern. Similar to Type 1, two types of vortices coexist in an alternating pattern; however, in Type 2, they form a vortex-antivortex relationship. The six domain states, $\alpha^{\pm}, \beta^{\pm}, \gamma^{\pm}$, have two possible phase shifts among $\{b^{\pm}, r^{\pm}\}$, $\{r^{\pm}, g^{\pm}\}$, and $\{g^{\pm}, b^{\pm}\}$, respectively. There is an additional constraint for $\Phi(L)$, the non-cyclic color order constraint: if the previous phase shift is r, then the allowed next phase shift is r^{-1}, b^{-1}, and g^{-1} is forbidden. If we cancel out the trivial backtracking paths, $\Phi(L)$ is always of the form $(rb^{-1}gr^{-1}bg^{-1})^n$. Typical examples are ferroic hexagonal manganites [9–13].

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